

Data Storytelling



Module 2: Creating Charts in Tableau

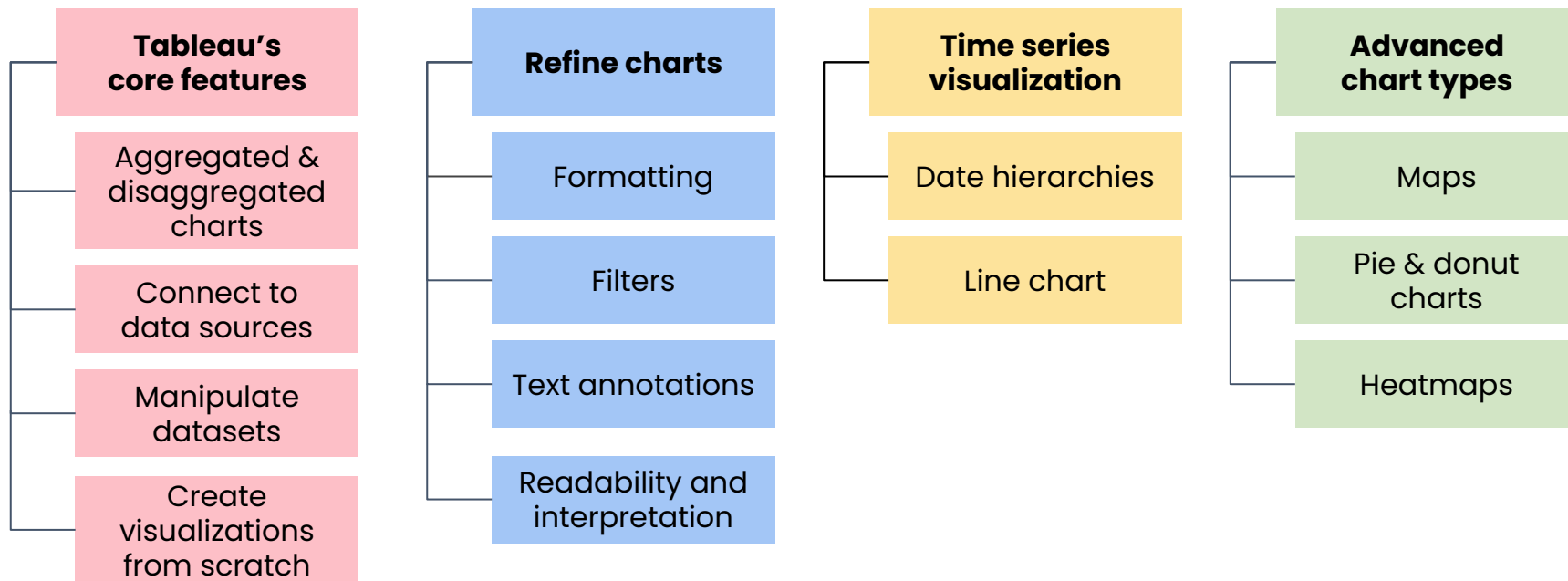




Creating Charts in Tableau

Module 2 introduction

Module 2 outline





Creating Charts in Tableau

Introduction to Tableau

What is Tableau?



- One of the most widely used tools in data vis
- Major advantages:

✓ **Connect to live data**

- Connect directly to live databases, cloud services, and APIs
- Visualizations update automatically as new data comes in

✓ **Create interactive visualizations**

- Users can explore data themselves
- Dynamic filters, tooltips, and links

✓ **Low code user interface**

- Can code up calculations
- Develop dashboards navigating user interface
- Speed advantage other than technical one
 - Can create simple linear regression
 - Need Python for multiple linear regression

Tableau Public vs. Desktop

In this course, you'll use:

- Free Tableau Public
- Limited functionality compared to Desktop
- Get core features for free



Tableau Desktop:

- Connect to wider variety of data sources
- Save your work offline
- Integrated AI feature: Tableau Agent



You'll need to create a free account to complete the labs for this course.

Available visualization software

- Common analogs:
 - Microsoft PowerBI
 - Google's Looker Studio
- Each have:
 - Slightly different interface
 - Ultimately accomplish similar tasks
- Skills in Tableau will directly transfer to these other programs

Creating Charts in Tableau

Creating a new workbook

Recap: Creating a new notebook

- ✓ Created a new workbook in Tableau Public
- ✓ Connected to data using a connector
- ✓ Explored tables and fields in Northwind database
- ✓ Published work to enable autosave feature

Dimensions

- **Qualitative:** text, dates, or discrete numbers (IDs)
- Used to segment and organize data
- Cannot be aggregated

Measures

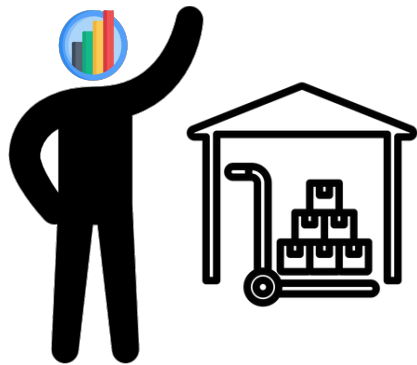
- **Quantitative:** sales, quantity sold, count of orders
- Numeric
- Can be aggregated



Creating Charts in Tableau

Setting up your data model

Scenario



You
Data Analyst



Company: Northwind Traders



Key goals:

- ☐ Evaluating product performance across different categories to see which items generate the most revenue.
- ☐ Analyzing sales by region to understand where most successful and where there might be room for growth.
- ☐ Identifying top customers based on factors like purchase frequency and total sales.

Recap: Getting to know your data

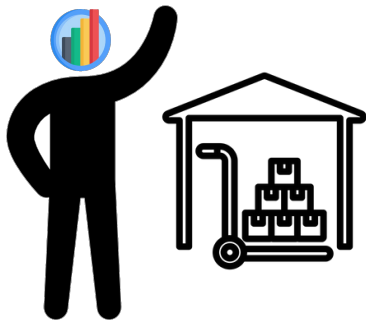
- ✓ Used **data source tab** to build a data model with only certain tables.
- ✓ Created relationships between tables by dragging and dropping them into workspace.
 - Tableau automatically creates relationships based on matching fields
 - You can manually select fields to match
- ✓ Switched to new worksheet
 - Triggering Tableau to build an extract containing only the essential tables



Creating Charts in Tableau

Creating aggregated charts

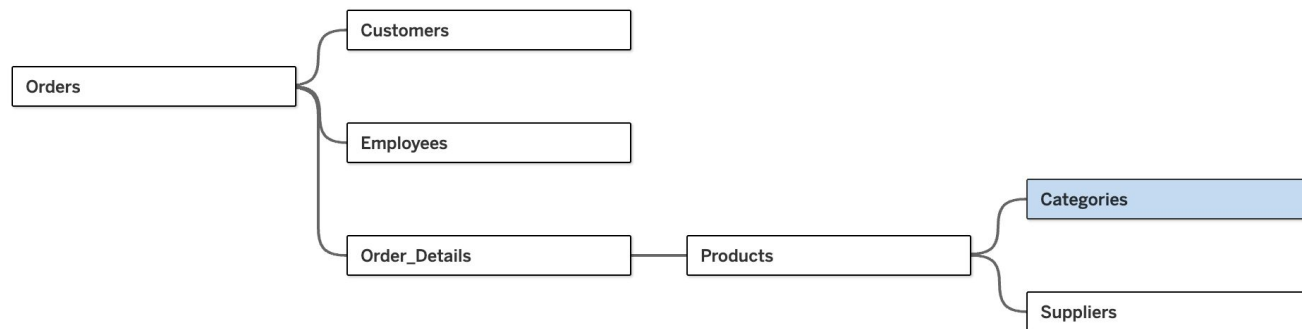
Scenario



You
Data Analyst

 **Company:** Northwind Traders

 **Data model:**



 **Goal:** Create visualizations to evaluate product performance, analyze sales by region, and identify top customers

Advice on creating charts

- Tableau's intelligent suggestions:
 - Provide a good starting point
 - Don't be afraid to experiment
- Pay attention to the aggregation settings
 - Default **sum** may not always be the best representation of data
 - Adjusting to **average** or **count** sometimes provide a more meaningful perspective.
- Use the Show Me panel to:
 - Explore different visualization options
 - Find the best chart type
 - Think about the relationship between dimensions and measures.

Recap: Creating aggregated charts

- Add dimensions and measures to the rows and columns shelves to create chart
 - Rows shelf: y axis
 - Columns shelf: x axis
- Use joined fields from other tables to segment
- Change summarization function for a chart using the dropdown measure for a field
- Explored different visualization options using the **Show Me** panel

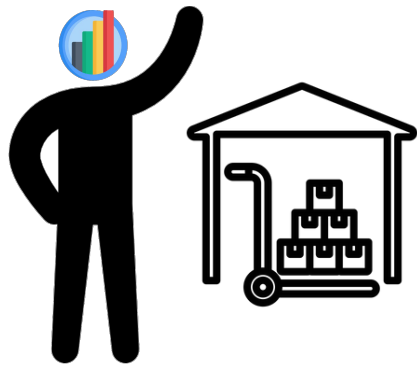
Aggregated chart

Combines multiple data points into single summary like sum or average

Creating Charts in Tableau

Calculated fields

Scenario



You
Data Analyst



Company: Northwind Traders



Completed Task: Grouped column chart comparing average units in stock and units on order by product category



Goal: Create visualizations to evaluate product performance, analyze sales by region, and identify top customers

Recap: Calculated fields

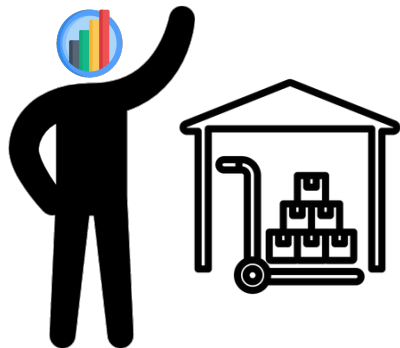
- Allow you to:
 - ✓ Create new fields based on other features
 - ✓ Define fields using custom computations
- Use custom programming language unique to Tableau
 - Hybrid between SQL and a spreadsheet
- Once calculated field is created
 - Drag new field into worksheet for visualization



Creating Charts in Tableau

Creating disaggregated charts

Scenario



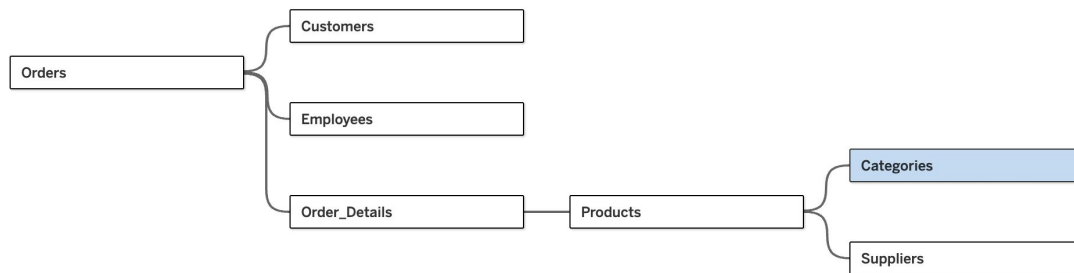
You
Data Analyst



Company: Northwind Traders



Data:



Goal: Create visualizations to evaluate product performance, analyze sales by region, and identify top customers

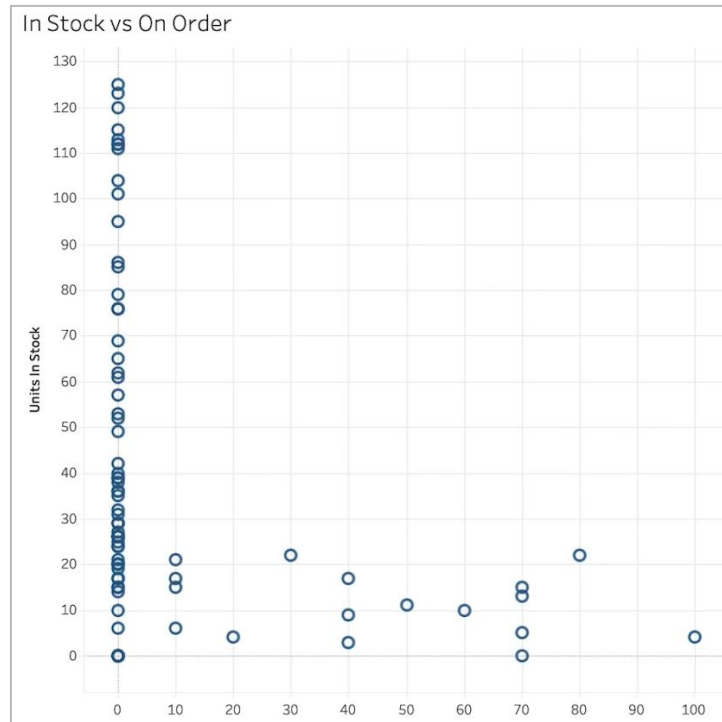


Task: Relationship between Units in Stock and Units On Order

- High number of units on order compared to its stock, indicate:
 - Shipping delays
 - Sudden surge in demand
 - Late stock orders that need to be placed

Recap: Creating disaggregated charts

- **Aggregated view:**
 - Identifying overall trends
 - Performance across categories
- **Disaggregated views:**
 - Reveal individual data points
 - Highlight variations and outliers
- You can:
 - Share workbook with others by closing edit view and copying link in browser
- Stakeholders can view or download workbook.



Creating Charts in Tableau

Data encoding

From customization to data encoding

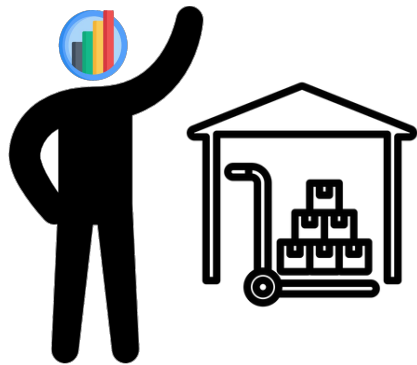
- Transform raw data into compelling stories using spreadsheets and Python
 - **Customizations:** color, marker size
- Draw attention to most important insights
- Data encoding:
 - Translating data into visual properties
 - Represent using color, size, shape or position
 - **Example:**
 - i. iPhone sales - ▲
 - ii. Android sales - ■



Chart elements:

- ☐ Labels
- ☐ Gridlines
- ☐ Titles
- ☐ Annotations

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create visualizations to evaluate product performance, analyze sales by region, and identify top customers



Visualize most popular products based on quantity sold **and** average unit price.



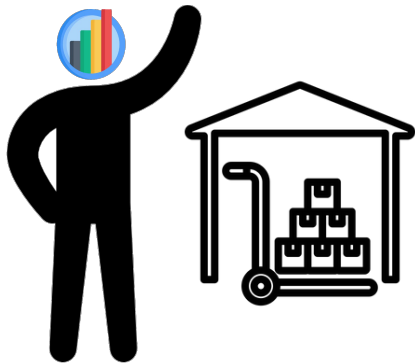
Improve functionality with better data encoding



Creating Charts in Tableau

Filtering & sorting

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create visualizations for a dashboard to evaluate product performance, analyze sales by region, and identify top customers.



Audience: Supply chain managers for different price ranges

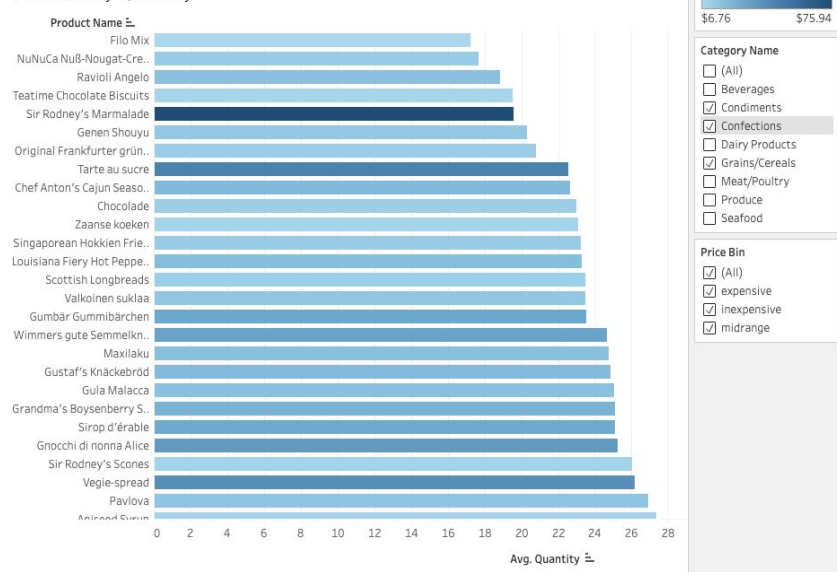


Interested in seeing data related to range they manage

Recap: Filtering & sorting

- Filtering lets you:
 - Narrow down view to focus only on the most relevant data
 - Specific product category
 - Particular numeric range
- Tableau allows you to:
 - Display filters so audience can select which ones are most relevant
 - Apply sorting to filtered visualizations
 - Apply multiple filters to single visualization

Products by Quantity



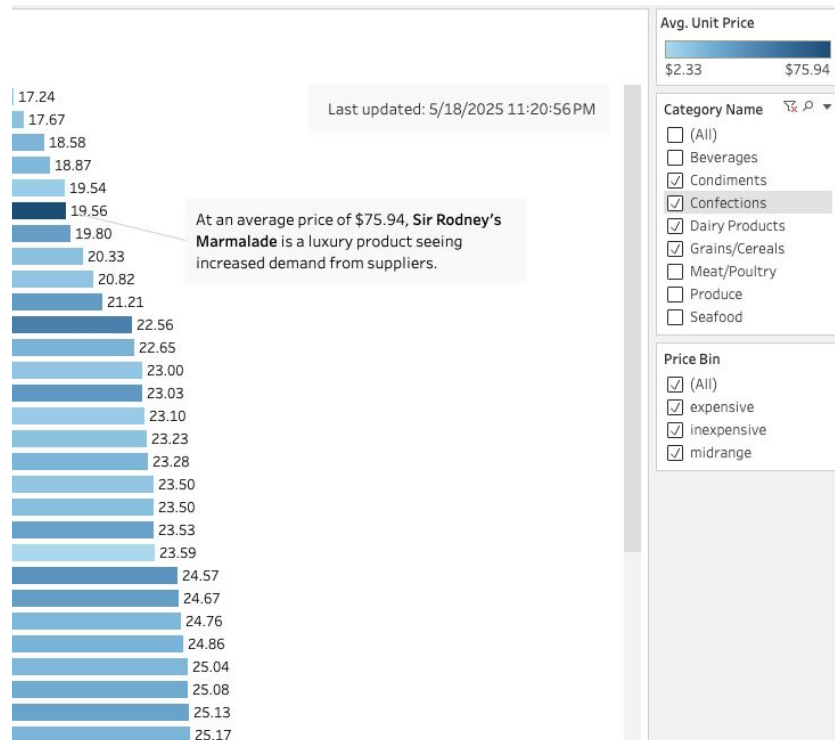


Creating Charts in Tableau

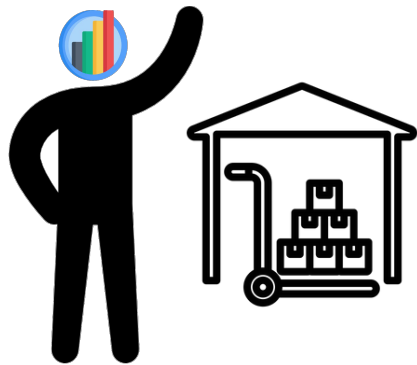
Labels and annotations

What are annotations?

- Improve clarity of any given chart:
 - Labeling axes
 - Providing chart title
 - Marker or bar labels
- Annotations:
 - Serve as powerful visual cues
 - Displaying main takeaways
- In Tableau:
 - Annotation color, text, and placement can be customized
 - Update dynamically as data changes



Scenario



You
Data Analyst



Company: Northwind Traders



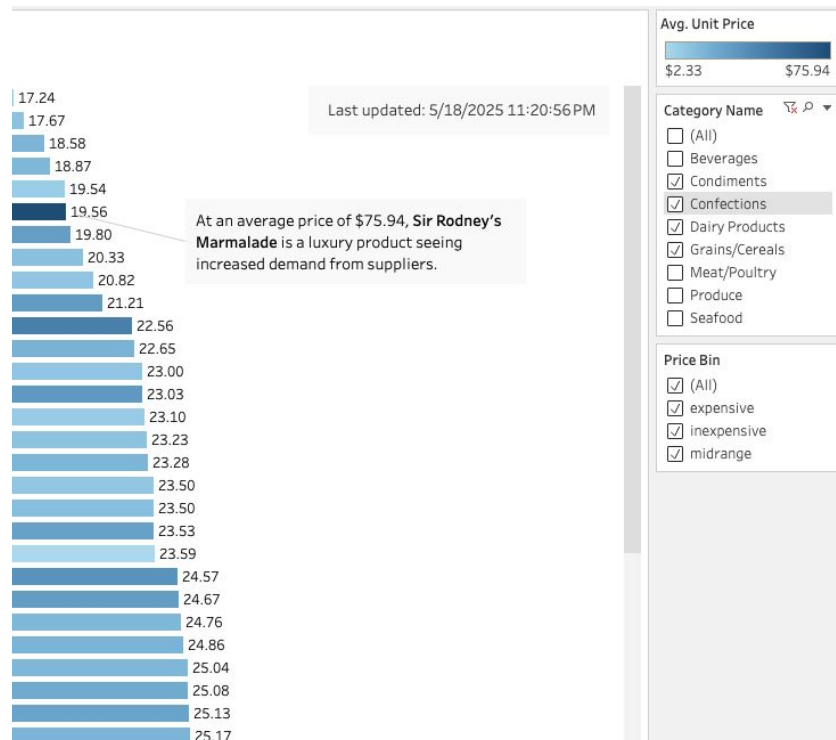
Status: Almost ready to share with supply chain managers



Working on a finishing touches to improve the clarity

Recap: Labels and annotations

- To edit titles and annotations in Tableau:
 - Use mix of static and dynamic content to create your annotation.
 - The **Insert** menu allows you to
 - Add fields like: `<data last updated>`
 - Use fields: `<Product Name>`
 - Use aggregations: `<AVG(Unit price)>`
 - Annotate either individual marks (e.g. bars), or the chart area itself
- Annotations are filtered dynamically

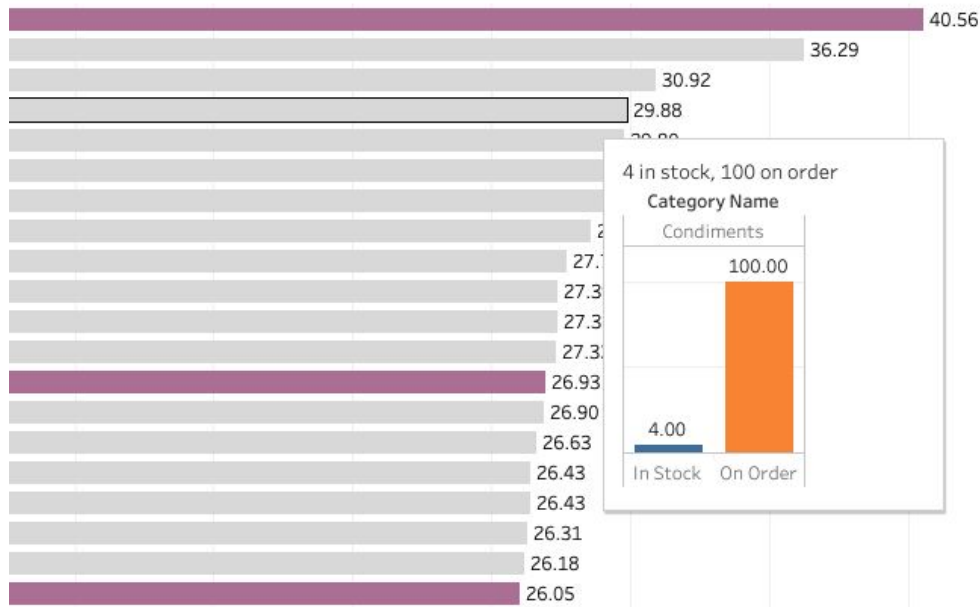


Creating Charts in Tableau

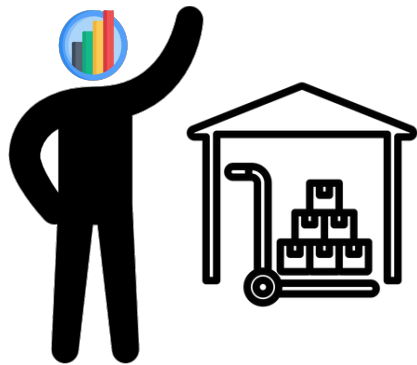
Tooltips

Tooltips

- Dynamic pop-ups
- Appear when the user hovers over a data point in visualization
- Provide additional context and insights without cluttering the main view
- Similar to annotations
 - Display static information and dynamic fields
- Use to reveal details on demand rather than crowding the main canvas
- Can add mini visualizations inside



Scenario



You
Data Analyst



Company: Northwind Traders



Task: Finalizing visualizations before sharing with supply chain managers



Create view highlighting certain confectionary products that are in high demand



Add additional on-demand info inside tooltips

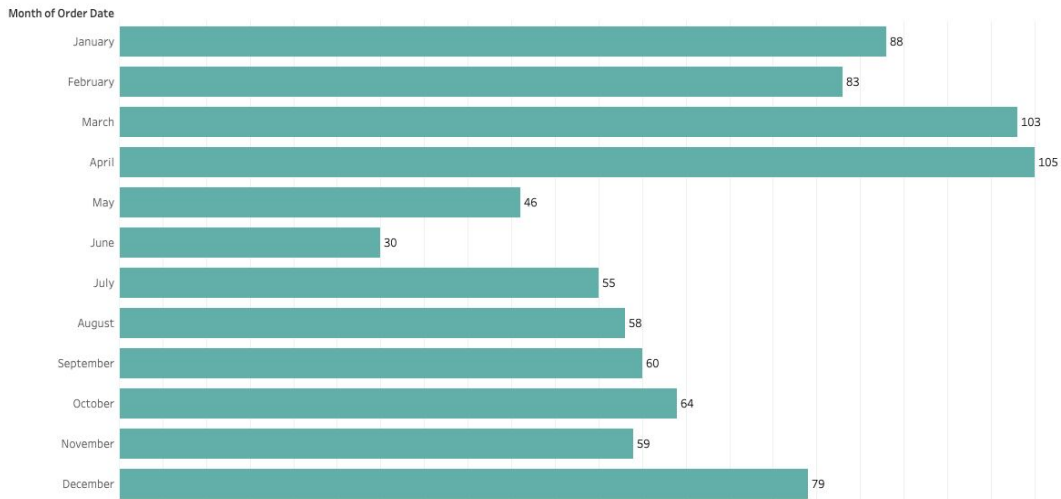
Creating Charts in Tableau

Dates & hierarchies

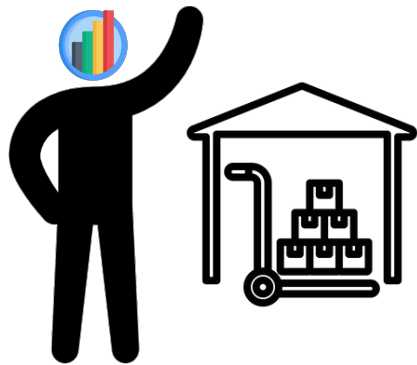
Dates and hierarchies

- Time series data:
 - **Earlier:** Used to visualize stock prices over time
- Process is similar in Tableau
 - Except dates come with built-in hierarchical relationships
 - Allow you to:
 - Analyze data at different levels
 - Used with other types of data

Orders by Month



Scenario



You
Data Analyst



Company: Northwind Traders



Task: Analyze order fulfillment for the shipping department at Northwind



Audience: Head of shipping department



Interested in developing new initiatives to improve order fulfillment speed

Recap: Dates & hierarchies

- Automatically created by Tableau if dates are well formatted
- Enable drill-down functionality
 - Users can expand from years to quarters, months, days, and hours.
- You can:
 - Apply filters and sorting to dates
 - Use calculated fields with dates
 - Subtracting dates
 - Using **DATEDIFF** to calculate time durations



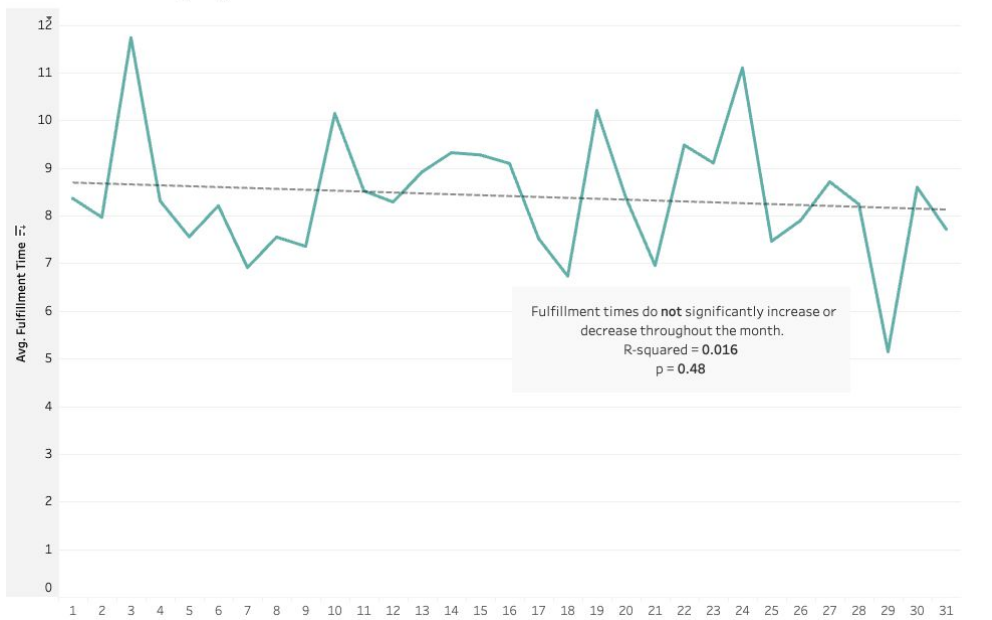
Creating Charts in Tableau

Visualizing time series

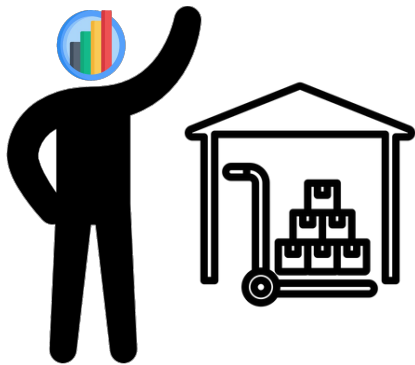
Line charts

- Used for analyzing time series data
 - Connecting points reveals changes over time more clearly
- Time series visualizations come with interactive features:
 - Trend lines
 - Forecasting
 - Dynamic date filters
- Derive insights from historical patterns and predict future behavior

Fulfillment Time by Day of the Month



Scenario



You
Data Analyst



Company: Northwind Traders



Task: Analyze order fulfillment for the shipping department at Northwind



Audience: Head of shipping department



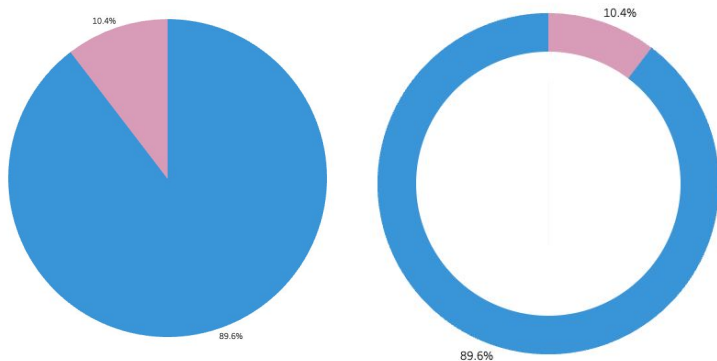
Believe fulfillment times increase towards end of the month, due to employees taking more time off

Creating Charts in Tableau

Pie & donut charts

Pie & donut charts

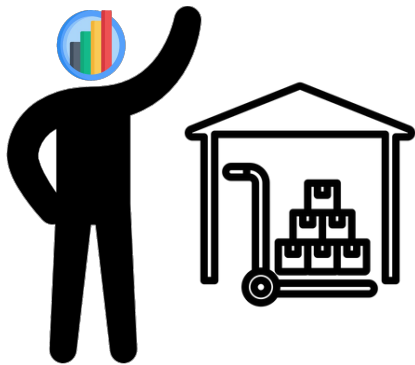
- ✓ Popular choices for visualizing part-to-whole relationships
- ✓ Work well when comparing relative sizes of small number of categories
- ✗ Can become visually overwhelming



Key best practices:

- Implement pie charts for part-to-whole relationships, rather than for comparing the absolute sizes
 - ☐ Consider whether a bar chart is better for your use case
- Use only 2 to 5 categories of data, and each slice is no less than 5% of the total.
- Reduce visual clutter in the chart:
 - ☐ Avoid 3D or drop shadows
 - ☐ Make data labels and legend clear
 - ☐ Colors are easily distinguishable

Scenario



You
Data Analyst



Company: Northwind Traders



Task: Create a new data product exploring item pricing for the Vice President of Sales



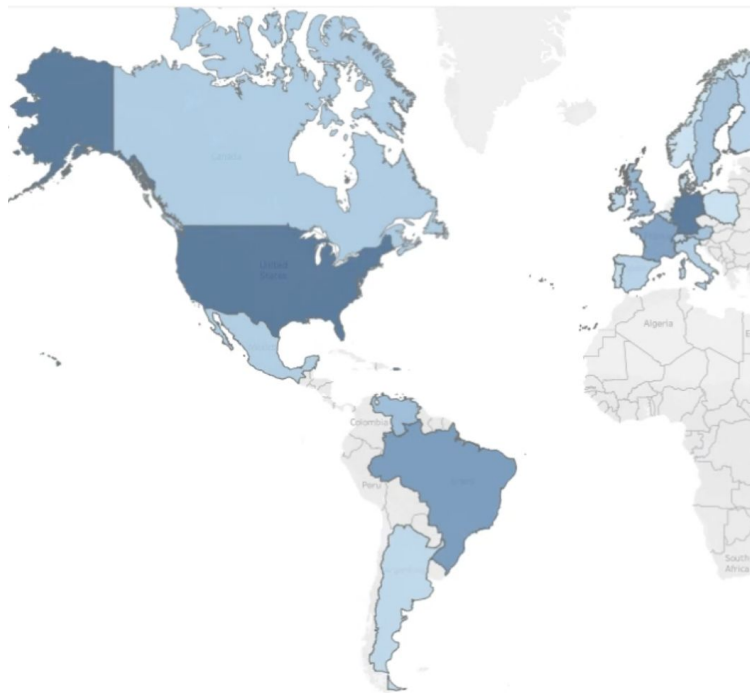
Goal: To better visualize the distributions of items of different prices and how well they are selling.

Creating Charts in Tableau

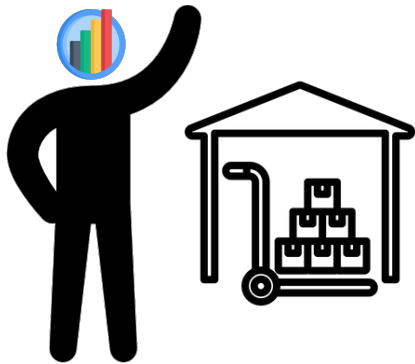
Maps

Maps

- Analysis involves multiple regions or countries
- Quickly see which areas stand out
- Examples:
 - Which customers are ordering the most
 - Where growth opportunities are the strongest
- Created with special geographic data types
 - Using latitude and longitude to position markers or shading



Scenario



You
Data Analyst



Company: Northwind Traders



Tasks: Create a new data product to better visualize:



Distributions of items of different prices



How well they are selling

Creating Charts in Tableau

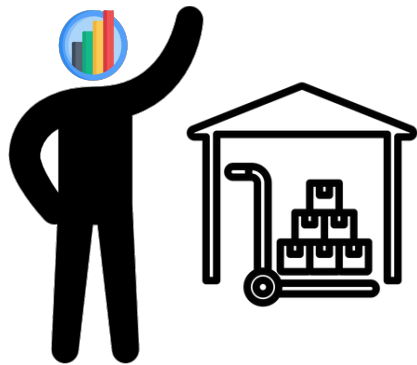
Tables

Tables

- Display and explore data in a familiar, spreadsheet-like format
- Organize information by rows and columns
- Interactive features:
 - Scrolling
 - Filtering
 - Dynamic sorting
- Present raw numbers
- Enhance with conditional formatting, calculated fields, and tooltips

Count...	Company Name	Beverages	Condiments	Confections	Dairy Products
Germany	Alfreds Futterkiste	33%	67%		33%
	Blauer See Delikatessen	29%	14%	43%	29%
	Die Wandernde Kuh	50%	10%	20%	60%
	Drachenblut Delikatessen	17%		17%	50%
	Frankenversand	40%	7%	47%	53%
	Königlich Essen	50%	29%	43%	43%
	Lehmanns Marktstand	40%	33%	33%	47%
	Morgenstern Gesundkost	20%	20%	60%	40%
	Ottilies Käseladen	30%	30%	40%	30%
	QUICK-Stop	54%	36%	36%	36%
USA	Toms Spezialitäten	50%		50%	17%
	Great Lakes Food Market	36%	9%	36%	36%
	Hungry Coyote Import Sto...			40%	20%
	Lazy K Kountry Store				50%
	Let's Stop N Shop	25%	75%	25%	
	Lonesome Pine Restaurant	50%		25%	25%
	Old World Delicatessen	30%	20%	50%	30%
	Rattlesnake Canyon Groc...	44%	22%	44%	56%
	Save-a-lot Markets	55%	16%	61%	48%
	Split Rail Beer & Ale	22%	11%	44%	33%
	The Big Cheese	25%	25%	25%	25%
	The Cracker Box	67%	33%		
	Trail's Head Gourmet Pro...	67%	33%	67%	33%
	White Clover Markets	50%	36%	43%	29%
Brazil	Comércio Mineiro		40%	20%	20%
	Familia Arguinaldo	57%		43%	29%
	Gourmet Lanchonetes	33%	22%	11%	33%
	Hanari Carnes	43%	43%	21%	43%
	Que Delícia	44%	56%	44%	22%
	Queen Cozinha	62%	31%	46%	46%
	Ricardo Adocicados	55%	9%	64%	36%
	Tradição Hinermercados	17%	33%	50%	50%

Scenario



You
Data Analyst



Company: Northwind Traders



Goal: VP of Sales expressed desire to explore the numbers of orders and customers by country.



Tasks: Put together a table of key metrics they can explore on their own

Creating Charts in Tableau

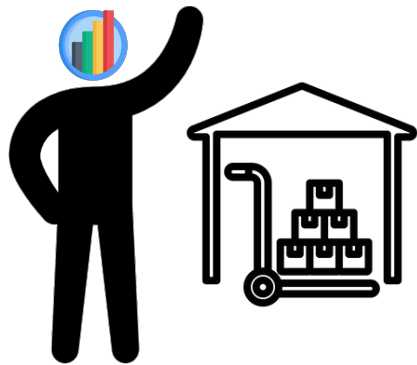
Heatmaps

Heatmaps

- Transform numerical data into vibrant color-coded cells
- Quickly spot patterns, trends, or problem areas in data.
- Especially useful to compare multiple categories or segments
- Most eye-catching spots indicate where you should focus attention



Scenario



You
Data Analyst



Company: Northwind Traders



Tasks: Put together a new visualization showing how product categories appear across orders in different countries



Goal: Present new data product to the VP of Sales to help them identify different markets to expand into